

Abelian Extensions Problem Set 4

1. A previous exercise asked you to show that $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \hookrightarrow (\mathbb{Z}/n\mathbb{Z})^\times$. Show that the map used in the first lecture is a surjection so $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.
2. Show that for each $m \in \mathbb{Z}^+$ there is a real number field K satisfying the following: $[K : \mathbb{Q}] = 2^m$, $K \subseteq \mathbb{Q}(\epsilon)$ for some root of unity ϵ , and 2 is the only prime ramified in K .
3. Show that there is only one real field K satisfying the conditions above by doing the following:
 - (a) Show that $\mathbb{Q}(\sqrt{2}) \subseteq K$.
 - (b) Show that $\text{Gal}(K/\mathbb{Q})$ must therefore be cyclic.
 - (c) Follow argument 4. given in the lecture for odd primes p to conclude.
4. Show that if $L/\mathbb{Q}$ is abelian with $[L : \mathbb{Q}]$ and δ_L both powers of 2, then $L \subseteq \mathbb{Q}(\zeta)$ for some n th root of unity ζ by showing that K , the maximal real subfield of the compositum $L(i) = L\mathbb{Q}(i)$ satisfies $L \subseteq K(i) \subseteq \mathbb{Q}(\zeta)$. (You first need to show that $[L(i) : K] = 2$.)